Morning

Properties of Regular Languages

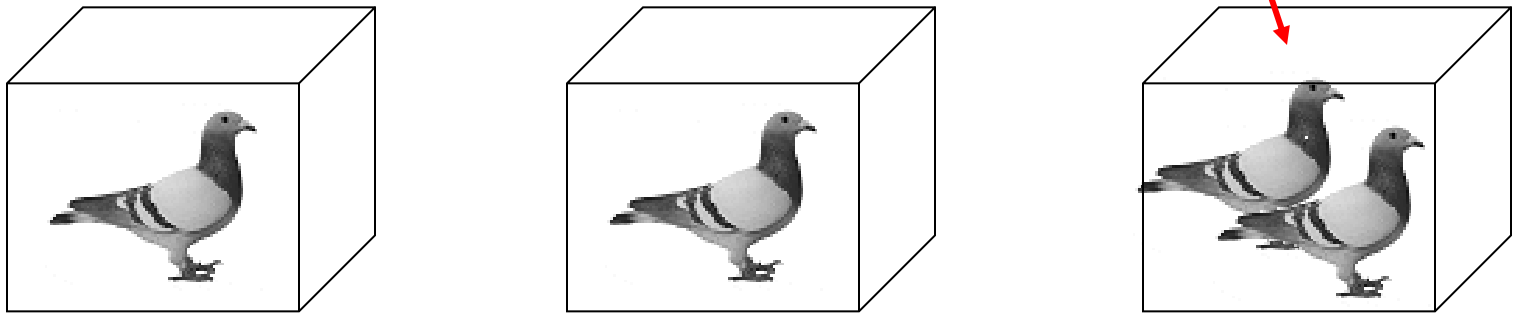
- Pumping lemma
- Closure properties

Pigeonhole Principle

4 pigeons

3 pigeonholes

A pigeonhole must contains at least two pigeons



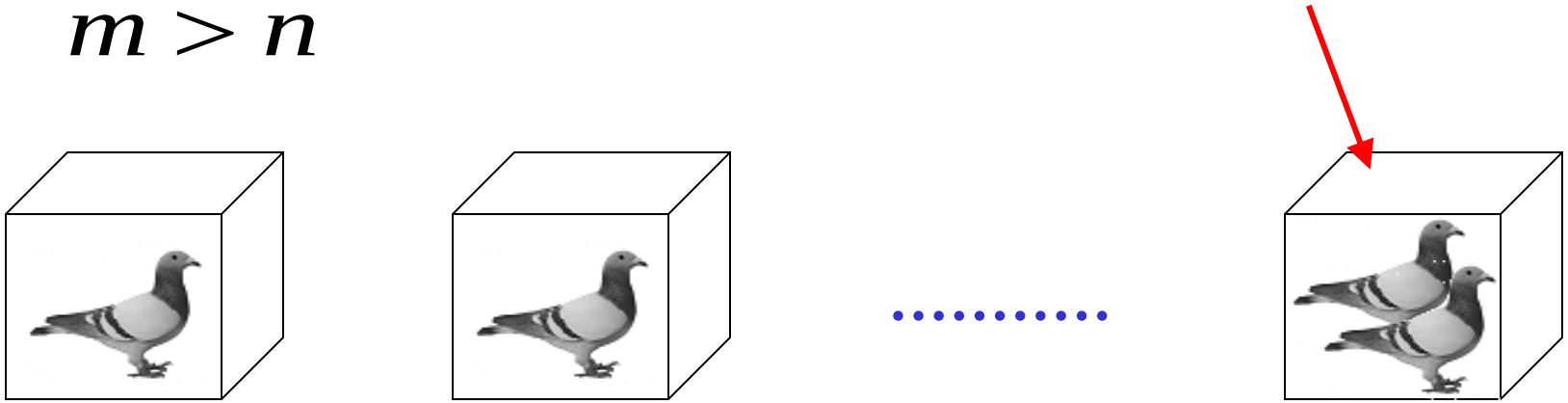
Pigeonhole Principle

m pigeons

n pigeonholes

$$m > n$$

There is a pigeonhole with at least 2 pigeons



DFA Principle

m symbols

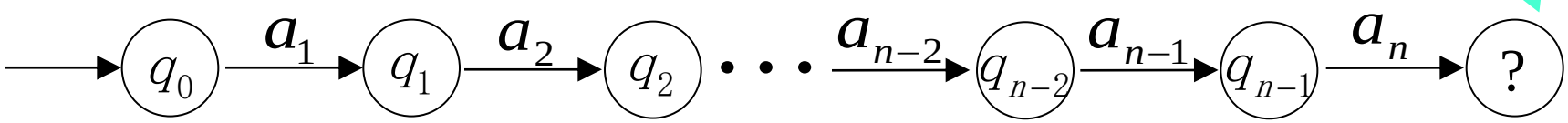
Let $A = (Q, \Sigma, \delta, q_0, F)$, and $n = |Q|$

$$w = a_1 a_2 \ldots\ldots a_m$$

n states

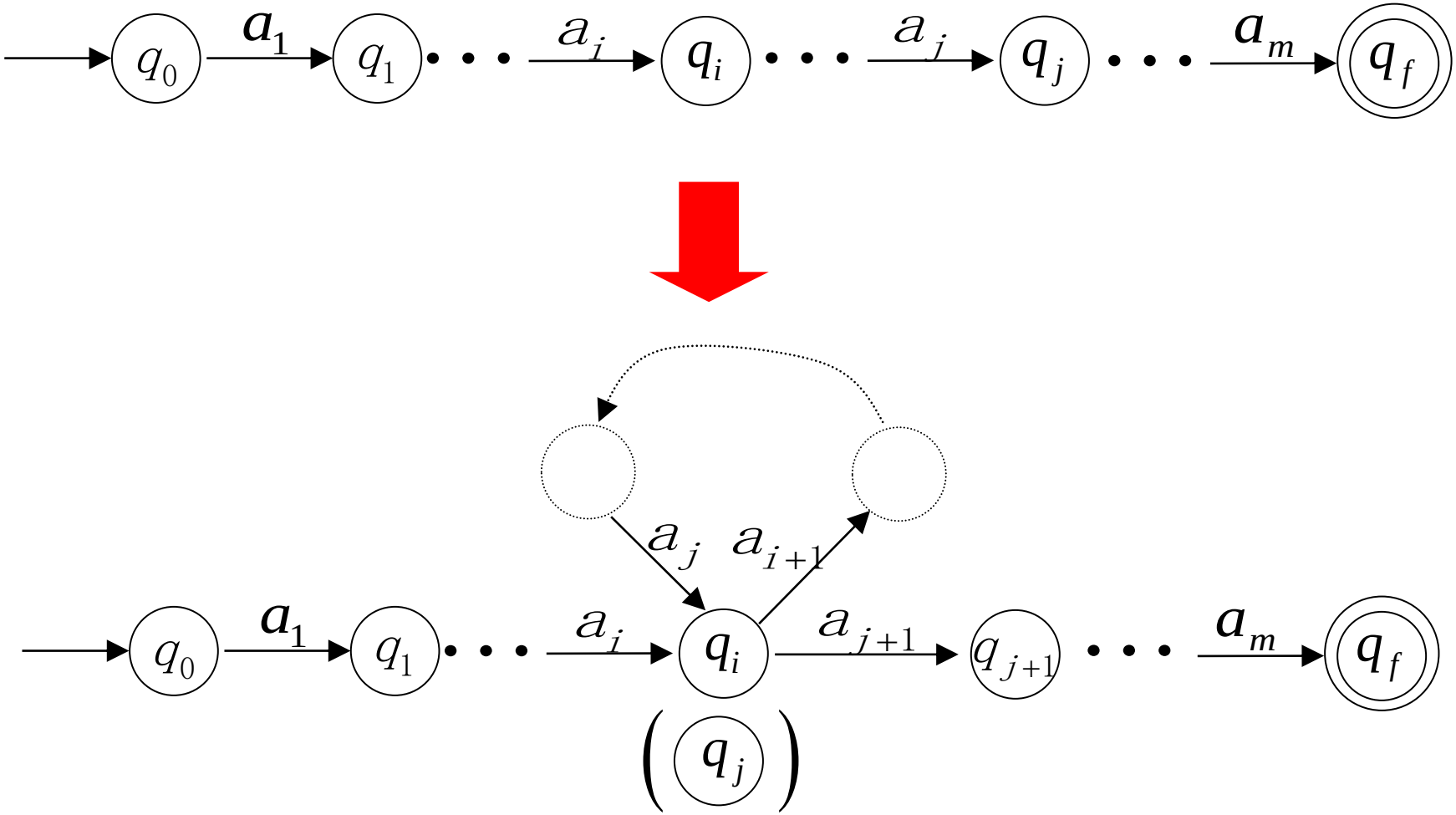
$$m \geq n \qquad a_n \cdots\cdots a_m \,?$$

$q_i \;:\; 0 \leq i \leq n-1$

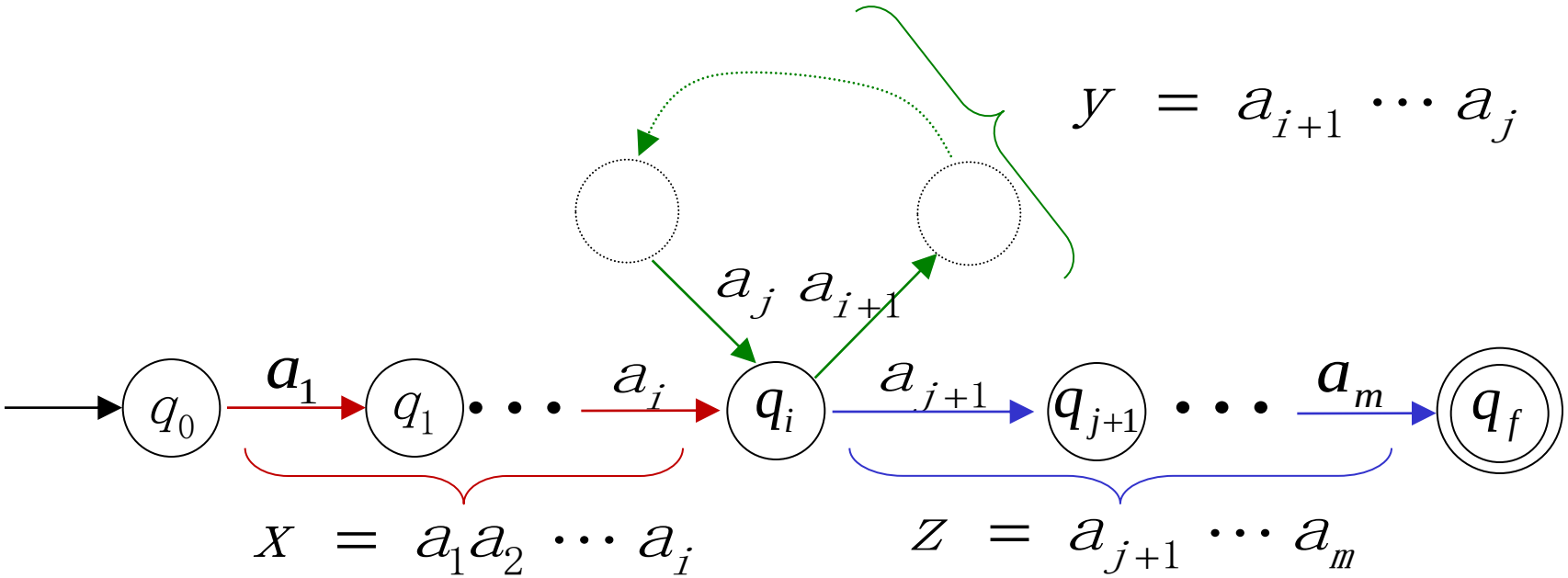


$$\exists q_i,\; q_j \Rightarrow q_i = q_j,\; 1 \leq i < j \leq n$$

DFA Principle



DFA Principle



$$\Rightarrow w = xyz \begin{cases} |xy| \le n \\ |y| \ge 1 \text{ or } y \ne \varepsilon \\ xy^k z \in L, \text{ for any } k \ge 0 \end{cases}$$

Pumping lemma

Pumping lemma for regular languages.

Let L be regular. Then

$\exists n, \forall w \in L : |w| \geq n \Rightarrow w = xyz$ such that

- $|xy| \leq n$
- $y \neq \varepsilon$ ($|y| >= 1$)
- $\forall k \geq 0, xy^k z \in L$

Decidable problem

Is *L* a regular language ?

Yes

- DFA
- NFA
- ε-NFA
- RegExp

No

- Pumping lemma

Example 1 "NO"

Let $L = \{ 0^n 1^n \mid n \geq 0 \}$. Is L regular ?

Suppose L is regular.

By pumping lemma there exist a constant n, for every $w \in L$, where $|w| \geq n$, w can be broken into three strings, $w=xyz$, such that $|xy| \leq n$, $y \neq \varepsilon$, and $xy^k z \in L$.

Get $w=0^n 1^n \in L$. Then $w=0^n 1^n = xyz$, and $xz=0^{n-|y|} 1^n \in L$.

It derived a contradiction(y contains at least one 0)

So L is not regular.

Example 2 "NO"

$$L = \{vv^R \mid v \in (a,b)^*\}$$

Get $w=a^n b^n b^n a^n \in L$.

for k=0, $xz=a^{n-|y|} b^n b^n a^n \in L$.

Example 3 "NO"

$$L = \{ 0^{n^2} \mid n \geq 0 \}$$

Closure properties

- union : $L \cup M$
- intersection : $L \cap M$
- complement : $\bar{L}$
- difference : $L - M$
- reversal : L^R
- closure(star) : L^*
- concatenation : LM
- homomorphism
- inverse homomorphism

Closure properties

- Union : $L \cup M$

Suppose $L(A)=L$, $L(B)=M$

Let $A = (Q_1, \Sigma_1, \delta_1, q_1, F_1)$, $B = (Q_2, \Sigma_2, \delta_2, q_2, F_2)$

$$C = (Q_1 \cup Q_2 \cup \{q_0\}, \Sigma_1 \cup \Sigma_2, \delta, q_0, F_1 \cup F_2)$$

$$\delta : \delta(q_0, \varepsilon) = \{q_1, q_2\}$$

$$\delta(q, a) = \delta_1(q,a), \forall (q, a) \in Q_1 \times \Sigma_1$$

$$\delta(q, a) = \delta_2(q,a), \forall (q, a) \in Q_2 \times \Sigma_2$$

Then $L(C) = L \cup M$

Closure properties

- Reversal $L^R = \{w^R \mid w \in L\}$

Convert $A(L)$ into $A(L^R)$ by :

- Reverse all the arcs of $A(L)$
- Convert start state of A(L) to accepting state of A(LR)
- Create a new state as start state of A(LR) with ε-transitions to all the accepting states of A(L)

Closure properties

- Reversal $L^R = \{w^R \mid w \in L\}$

Suppose $L(A)=L$ where A is a DFA

Let $A = (Q_1, \Sigma, \delta_1, q_1, F_1)$, $B = (Q_2, \Sigma, \delta_2, q_0, \{q_1\})$

$$Q_2 = 2^{Q1} \cup \{q_0\} \quad (q_0 \notin Q_1)$$

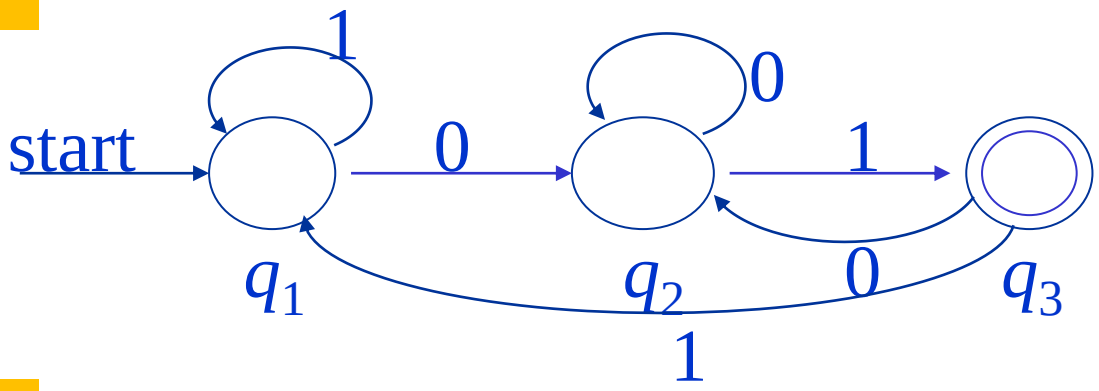
$$\delta : \delta(q_0, \varepsilon) = \{q \mid q \in F_1)\}$$

$$\delta_2(q, a) = \{ p \mid \delta_1(p, a) = q \}$$

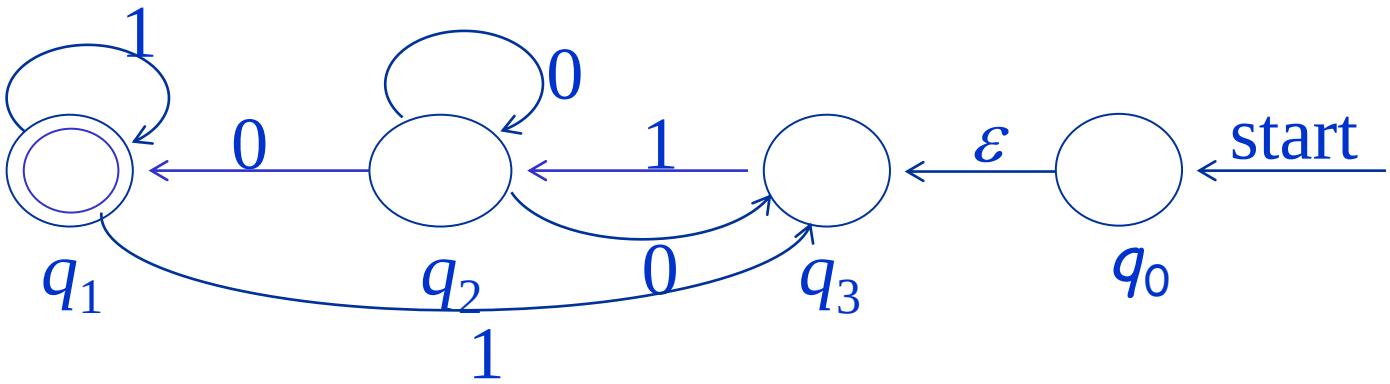
Then $L(B) = L^R$

Example 3 Convert closure

(0+1)*01



10(0+1)*



Closure properties

- Complement

$$\overline{L} = \{w \mid w \in \Sigma^* \text{ and } w \notin L\}$$

Let DFA A=(Q, Σ, δ, q_0, F) , and $L(A)=L$

Let DFA B=(Q, Σ, δ, q_0, S) , and S=Q-F

Then L(B)= $\overline{L}$

Closure properties

- Intersection : $L \cap M$

$$\overline{\overline{L} \cup \overline{M}}$$

Suppose $L(A)=L$, $L(B)=M$

Let $A = (Q_1, \Sigma, \delta_1, q_1, F_1)$, $B = (Q_2, \Sigma, \delta_2, q_2, F_2)$

$$C = (Q_1 \times Q_2, \Sigma, \delta, (q_1, q_2), F_1 \times F_2)$$

$$\delta : (Q_1 \times Q_2) \times \Sigma \rightarrow Q_1 \times Q_2$$

$$\delta((p, q), a) = (\delta_1(p,a), \delta_2(q,a))$$

Then $L(C) = L \cap M$

Homomorphism

$h: \quad \Sigma^* \rightarrow \Gamma^*$

Let $w=a_1a_2\ldots\ldots a_n \in \Sigma^*$, then

$h(w)=h(a_1)h(a_2)\ldots\ldots h(a_n)$

Let $\Sigma=\{ 0, 1 \}$, $\Gamma=\{ a, b \}$, $h(0)=ab$, $h(1)=\varepsilon$

$h(0110)=h(0)h(1)h(1)h(0)=ab\varepsilon\varepsilon ab=abab$

$h(L)=\{ h(w) \mid w \text{ is in } L \}$

Homomorphism

Regular language is closed under homomorphism.

Assume r is a *RegExp* and $L=L(r)$.

For any symbol a of r, h(a) is a *RegExp*

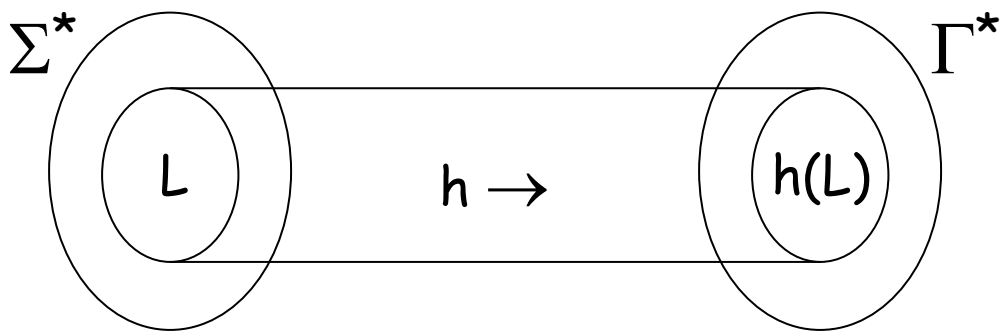
$\Rightarrow$ h(r) is *RegExp*.

$\Rightarrow$ $h(L) = L(h(r))$ is *RegLang*.

Inverse homomorphism

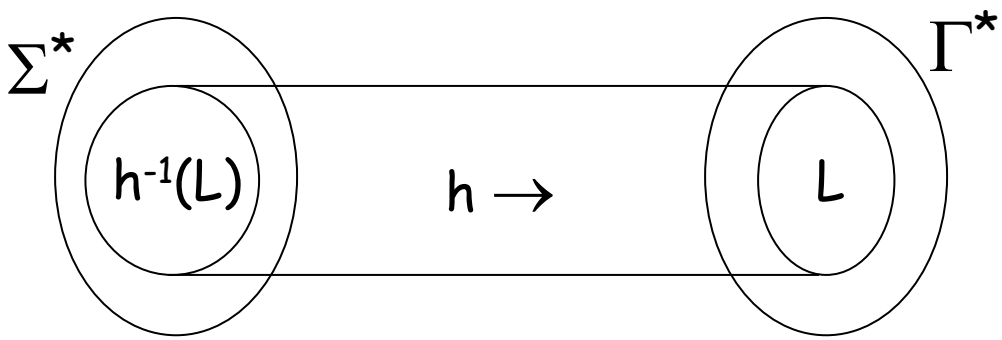
$h: \quad \Sigma^* \to \Gamma^*$

$h^{-1}(L)=\{ w \mid h(w) \text{ is in } L \}$



$\forall w \in L \Rightarrow h(w) \in h(L)$

$\forall v \in h(L)$
$\Rightarrow \exists w \in L : h(w)=v$



$\forall w \in h^{-1}(L) \Rightarrow h(w) \in L$

$\forall v \in L \Rightarrow$
$\exists w \in h^{-1}(L) : h(w)=v$?

Example 4 Inverse

Let Σ={ a, b }, Γ={ 0, 1 }, h(a)=01, h(b)=10

Let L = $\{00,1\}^*$ then $h^{-1}(L)$ = ?

L ={ ε,1,00,11,100,001,0000,111,1100,1001, 0011,

10000,00100,00001,000000,1111,11100,11001,... }

$h(\{a,b\}^*) = \{01,10\}^*$

={ε, 01,10,0101,0110,1001,1010,010101,010110,

011001,100101,011010,100110,101001,101010,... }

Example 4

Let Σ={ a, b }, Γ={ 0, 1 }, h(a)=01, h(b)=10

Let L = {00,1}* then $h^{-1}(L)$ = ?

{ 00,1 }* $\cap$ { 01,10 }*

={ε, 1001,10011001,100110011001,..... }

h(aa)=0101, h(ab)=0110, h(ba)=1001, h(bb)=1010

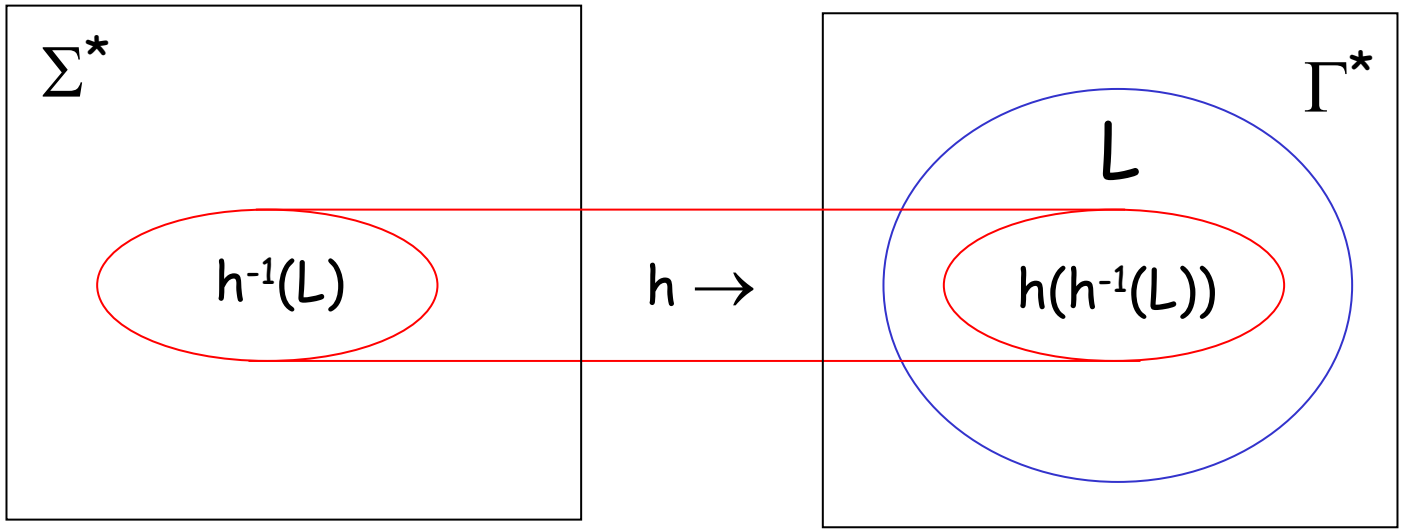
h({ba}*) = {1001}* $\subset$ { 00,1 }*

$\Rightarrow$ $h^{-1}(L)$ = {ba}* = { w | h(w)$\in L$ }

Example 4

Let $\Sigma=\{ a, b \}$, $\Gamma=\{ 0, 1 \}$, $h(a)=01$, $h(b)=10$

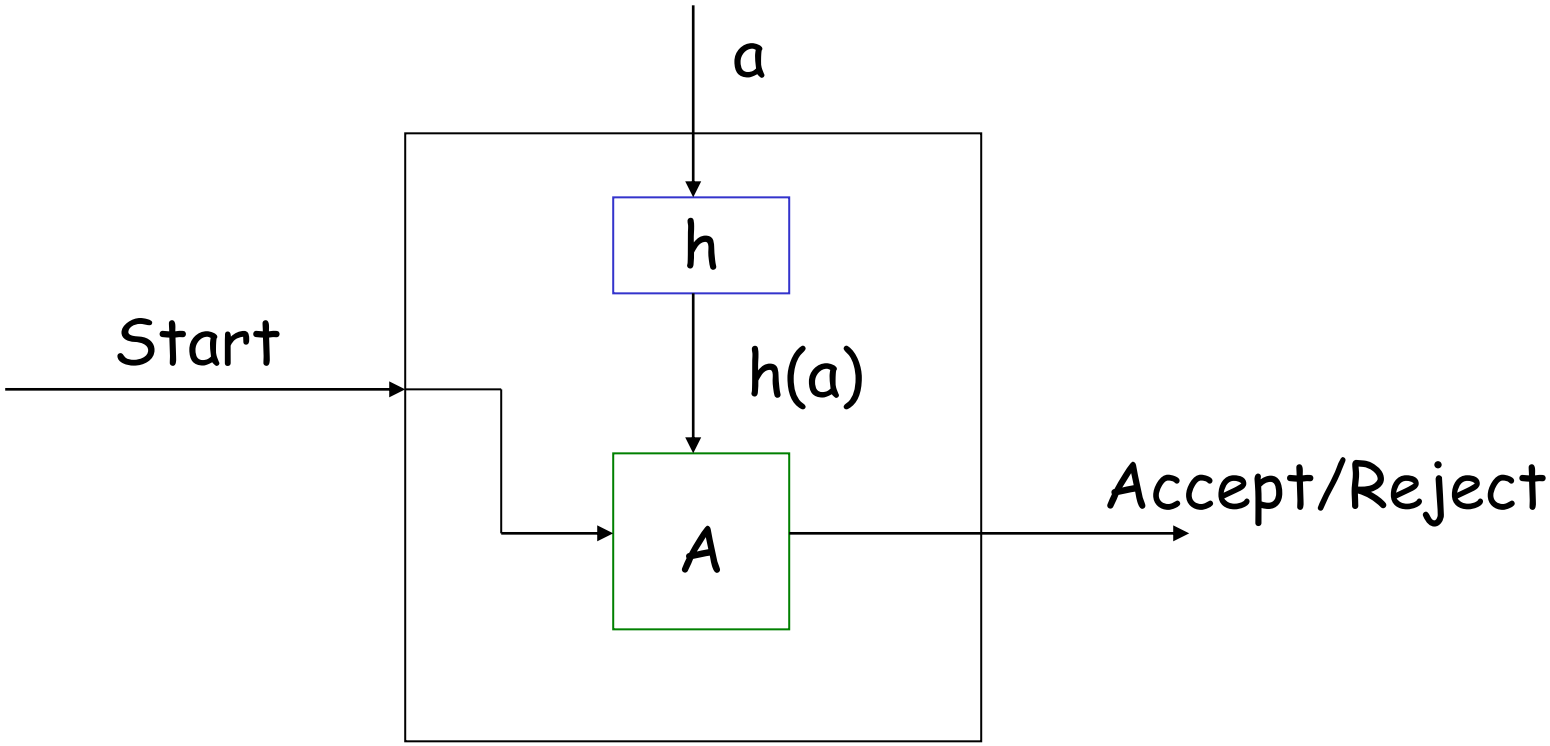
Let $L = \{00,1\}^*$ then $h^{-1}(L) = \{ba\}^*$



$h^{-1}(L)=\{ba\}^*$, $h(h^{-1}(L))=\{1001\}^* \subset L=\{00,1\}^*$

Inverse homomorphism

RegLang is closed under inverse homomorphism.



$$A = (Q, T, \delta, q_0, F), \qquad B = (Q, \Sigma, \gamma, q_0, F)$$

$$where \;\; \gamma(q, a) = \hat{\delta}(q, h(a))$$

Good good study day day up !